Abdullah Khan

437-488-8659 | abdullah.khan1@uwaterloo.ca | khanzai.vercel.app | linkedin.com/in/khanzai | github.com/khanuzai

Education

University of Waterloo — Waterloo, ON
Bachelor of Computer Science (BCS) — *Expected 2029*

Wilfrid Laurier University (Double Degree) — Waterloo, ON
Bachelor of Business Administration (BBA) — *Expected 2029*

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, SQL, C/C++, Racket
Frameworks & Libraries: React, Node.js, Express.js, FastAPI, Puppeteer, Tailwind CSS, Material UI
Tools & Technologies: Git/GitHub, GitLab, PostgreSQL, MongoDB, Docker, AWS (Lambda, S3, RDS, ECS), Grafana, Graylog, Vercel, JIRA, Figma

Experience

Software Engineering Intern — Aurora, ON
LeapAP Inc. — *May 2026 – Aug. 2026*

- Built **2 parsers** and **3 scrapers** integrating **5 billing portals** and processing **20+ invoices** in production within the first week of onboarding.
- Resolved **2 triage incidents** by debugging scraper failures using **Grafana** and **Graylog**, identifying root causes such as bot detection blocks and malformed HTML structures.
- Shipped **20+ commits** to production via **GitLab**, contributing to a high-velocity deploy pipeline that pushes to prod multiple times daily.
- Engineered bot detection circumvention logic and invoice parsing pipelines using **Node.js** and **Puppeteer**, handling edge cases across portals with varying layouts and anti-scraping measures.

Software Engineering Intern — Milton, ON
PixelsBoost — *Sep. 2025 – Dec 2025*

- Delivered **5 full-stack client websites** to production using React, HTML/CSS, and JavaScript, implementing dynamic contact forms, interactive image galleries, and mobile-responsive navigation serving **2,000+ monthly users**.
- Architected and integrated **3 third-party APIs** including Stripe for payment processing (**$15K+ monthly transactions**), Google Maps for geolocation, and SendGrid for email automation, maintaining **99.5% uptime** through error handling.
- Boosted website performance by **42%** through WebP image compression, lazy loading, and Cloudflare CDN implementation, improving Google Lighthouse scores from **68 to 87** and reducing bounce rate by **18%**.
- Managed **5 concurrent client projects** using Git/GitHub, making **150+ commits** across **25+ feature branches** while collaborating with 2 designers and resolving **12+ merge conflicts**.

Projects

Attack Surface Growth Simulator (ASGS) [GitHub] | *Python, React, FastAPI, SQLAlchemy, NumPy, SQLite, Recharts, Pydantic*

- Built full-stack security risk modeling tool that calculates attack surface scores (**0-100**) across **5** threat categories by normalizing system metrics (endpoints, users, MFA adoption, vulnerabilities) and generating ranked driver breakdowns.
- Developed **FastAPI** backend with **Pydantic** validation that computes quadratic risk functions and calculates derivatives using **NumPy** to identify unsafe growth zones where system complexity creates disproportionate security exposure.
- Designed **SQLite** database with **SQLAlchemy** ORM to persist assessment configurations and results, enabling users to compare different security scenarios and track how architectural changes impact overall risk posture.
- Built interactive **React** dashboard with **Recharts** that visualizes risk curves, growth rates, and danger zones in real-time as users adjust system parameters, helping teams identify critical inflection points before scaling.

BookPulse [GitHub] | *React, TypeScript, FastAPI, PostgreSQL, Docker, AWS (Lambda, S3, RDS, ECS, EventBridge)*

- Built and deployed full-stack **real-time book price intelligence platform** with React/TypeScript frontend on Vercel and FastAPI backend on Render, connected to **AWS RDS PostgreSQL** tracking **100+ books** with live rarity and demand scoring.
- Architected serverless **AWS Lambda** data ingestion pipeline triggered by **EventBridge** on an hourly schedule, storing raw book data in **S3** and computing price change metrics, demand scores, and rarity rankings across **5 categories**.
- Containerized backend with **Docker** and deployed to **AWS ECS Fargate**, configuring **IAM** roles, security groups, and **CORS** middleware to enable secure cross-origin communication between Vercel frontend and cloud backend.
- Implemented animated real-time dashboard with **Chart.js** visualizations, top movers table, and rarity leaderboard, resolving a **React StrictMode** animation bug caused by double-invoked effects corrupting `useRef` state on mount.